Taxonomy-Based Lightweight Cryptographic Frameworks for Secure Industrial IoT: A Survey

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***Abstract*—Manufacturing and automation are undergoing a radical transformation thanks to the advancement of the Industrial Internet of Things (IIoT), which enables real-time tracking, predictive analysis, and interconnected device ecosystems. However, the heterogeneous and massive deployment of devices deprives IIoT systems of a security advantage. Traditional cryptographic methods, while secure, are prohibitively computationally expensive for such settings. lightweight cryptography models are promising alternatives in terms of providing quality security with low resource consumption. In this article, we propose a structured classification of lightweight cryptographic frameworks for IIoT depending on the types of algorithms, functional goals, deployment domains, and application domains. Comparisons reveal the computation performance, energy, and memory costs among different methods, and may provide some hints for practitioners to determine the best deployment. In contrast to the previous works, in this article, we consider the security challenges of the IIoT and derive a layered framework for performance-/application-based selection. Future trend directions highlight the search for post-quantum cryptographic solutions, scalable authentication, and energy-aware hybrid approaches, anchoring this survey as a roadmap for the secure development of IIoT.**

***Index Terms*—Cryptographic taxonomy, Industrial Internet of Things (IIoT), lightweight cryptography (LWC), resource-constrained devices, security in IIoT.**

I. Introduction

IN TODAY'S industrial systems, the Industrial Internet of Things (IIoT) represents a revolutionary approach. It allows real-time monitoring, predictive maintenance, and automation. IIoT improves operational efficiency and optimizes industrial processes by integrating smart devices, sensors, and communication networks [1], [2]. Security is a big issue, however. Because of the number of devices working together and the sensitive nature of their data, there are some serious problems. Cyberattacks, such as data breaches, unauthorized access, and Denial of Service (DoS) attacks can all interrupt critical operations within an industry; combined with cost losses and damage to safety, they leave little room for error [3].

While traditional cryptographic methods are highly secure, they can be resource-intensive, and thus not a good match for the constraints of IIoT devices [4]. These devices usually have only a small amount of computational resources, memory, and energy at their disposal, so a cryptographic technique scaled down in complexity is needed too. Lightweight cryptography (LWC) finds the right balance between security and resource consumption for IIoT systems; nowadays it is a must [5], [6].

However, despite its importance for IIoT, LWC is still very fragmented, and there is not unified framework available to classify or evaluate the technologies being proposed [7], [8]. As a result, many particularly worthwhile approaches have been overlooked, and many unsuitable ones tested.

The survey is aimed at plugging this gap in knowledge by providing a comprehensive framework that can be used to classify and assess lightweight cryptographic systems meant for IIoT. We divide cryptographic techniques in all their forms by: 1) functions or criteria; 2) types of algorithms (ranging from Symmetrical to Elliptical); 3) the field in which they are put to use; 4) the domain the application is intended for; and most importantly 5) energy consumption and efficiency. But we also consider such performance metrics as they exist, computational efficiency, memory requirements, etc, together before deciding what might be best for you when researching lightweight cryptographic methods.

With this survey, we make several important contributions to the field of LWC for IIoT.

1) *Categorization:* We categorize lightweight cryptographic schemes according to their cryptographic function, algorithm type, target implementation domain, and application domain. The taxonomy fosters understanding and aids in systematic method selection for different IIoT cases by considering these various factors in a comprehensive way.
2) *Quantitative Comparison:* A quantitative comparison with three state-of-the-art works illustrates that our proposed scheme is unique in tackling such IIoT-specific

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TABLE I
QUANTITATIVE COMPARISON OF OUR SURVEY WITH RECENT WORKS

Metric	Our Survey	[9]	[10]	[11]
Coverage of Cryptographic Techniques	Satisfied	Partially Satisfied	Satisfied	Not Satisfied
Depth of Analysis	Satisfied	Partially Satisfied	Partially Satisfied	Not Satisfied
Taxonomy-Based Organization	Satisfied	Not Satisfied	Partially Satisfied	Not Satisfied
Focus on IIoT-Specific Challenges	Satisfied	Satisfied	Partially Satisfied	Not Satisfied
Research Gaps and Future Directions	Satisfied	Partially Satisfied	Not Satisfied	Not Satisfied
Novel Contributions and Emerging Secure IIoT Directions	Satisfied	Not Satisfied	Not Satisfied	Not Satisfied

security challenges. Unlike previous surveys, this article includes state-of-the-art contributions (2020–2025) and adopts a taxonomy-based methodology to organize the literature, which highlights its novelty and importance.

3) *Performance Analysis:* The survey provides a detailed description of the performance of light-weight cryptographic operations based on computational efficiency, energy efficiency, and memory requirement. This work provides useful recommendations on how to choose a cryptographic solution according to the context in IIoT.
4) *Identifying Research Gaps:* This survey highlights key gaps—not much adoption of post-quantum cryptography (PQC), challenges rolled out during scalability of security for IIoT systems, and lack of standardization—which need to be addressed to further the development of secure IIoT systems by referring to the recent literature along with latest techniques.
5) *Novel Contributions and Fledging Secure IIoT Trends:* This article sets forth a layered application-oriented selection framework that serves as a bridging link between theoretical design and pragmatic deployment of cryptographic systems in IIoT. It highlights future lines of research, such as scalable authentication methods, post-quantum secure solutions, and energy-aware hybrid implementations that are essential in the design of robust and adaptable IIoT infrastructures.

The remainder of this article is arranged as follows. Section II provides a quantitative comparison of this survey with three recent survey studies. Section III will give an introduction to IIoT, the security challenges that it brings with it, and an overview of LWC. Section IV puts forward the taxonomic approach to lightweight cryptographic techniques. Section V makes a stats competition and comparative analysis of these techniques spoken about. Section VI identifies the gaps in research and sketches future research directions. Section VII proposed novel contributions and emerging secure IIoT directions. Last, Section VIII concludes this article with main findings and suggestions for further work.

II. QUANTITATIVE COMPARISON

The part highlights the peculiarities and superiority of this survey by means of comparing it with three recent survey papers on the subject. This research [9] discusses the intuitive fusion of blockchain and LWC, which is mainly in order to enhance IOT and IIOT security. It reviews the possible applications of various encryption algorithms in an IOT intrusion detection system. This survey [10] classifies cryptographic protocols for IOT and IIOT based on the blockchain, with a focus on scalability, energy saving, as well as security relationship costs. This article [11] thoroughly reviews interoperability challenges and cryptographic standards for IIOT. By introducing a taxonomy that covers lightweight cryptographic applications, it is intended to provide some guidance. Comparisons were made along five key metrics that measure the extent to which a survey is comprehensive and useful to practitioners.

1) *Comparative Metrics:* In each case, this survey is rated for this particular key quality of a standard review.
2) *Coverage of Cryptographic Techniques:* This measures fittingly the wider breadth of cryptographic methods that have been integrated, but not whether or not such a method has ever been applied in an industrial field.
3) *Analysis Depth:* The survey does indeed provide technical insights, performance metrics, tradeoffs reflecting on each technique's advantages and minor flaws, etc.
4) *Remembering Them by Name:* The presentation and detailed explanation of the taxonomy, whether or not the survey has actually tried to set out a systematic structure, according to which it can overcome these challenges.
5) *Stepping into IIoT:* Whether the survey addresses some special IIoT requirements like energy conservation, real-time communication, and how different systems work together.
6) *Research Gaps and Future Directions Analysis:* Whether the survey has put forward some practical proposals about what remains to be done for LWC within the next few years
7) *Quantity of Recent Studies Reviewed:* The greater the reliance placed upon older articles, the less space left for recent, breaking stories as they happen on academic journals and in-house technical conference reports.

As listed in Table I, we conclude the following: This survey comprehensively researches lightweight cryptographic techniques, involving both symmetric and asymmetric methods, hash functions, and applications for specific application domains. The second problem is that the content of Survey [9] is incomplete in accomplishing this goal, and Survey [11] gives only rough estimates across these two sections surveyed. Unparalleled in an earlier study of LWC techniques, this project further includes a discussion on measures, such as computational efficiency, energy consumption, and time requirements. It offers some insights into the process of making tradeoffs between various factors. Surveys [10] and [11] do not do so at all from this point of view. What is particularly innovative in this survey project is its structured taxonomy. Techniques are listed by their algorithmic type and functional objective, for example, along with which domain

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they are used in and under what circumstances. Our surveys have yet to make a comprehensive, systematic classification like the one provided here. A major achievement of this research is to address specific needs in terms of energy efficiency, scalability, and real-time performance for IIoT. Although Survey [9] makes an interim response to such needs, Surveys [10] and [11] are nonspecific at best; if anything specific, it would be setting them aside as being even farther from our intentions today. By identifying specific deficiencies and proposing practical directions for the future, this survey provides a program to carry out further research. Other surveys often overlook this aspect and therefore have little impact on the research community. This survey includes the most recent research results, thus keeping its relevance. It would be a pity if Survey [11] were to cite some long-outdated articles in order to address current IIoT challenges.

In contrast to this, our proposed categorization introduces a layered, integrated framework that classifies lightweight cryptographic algorithms along five major dimensions: 1) cryptographic function; 2) algorithmic class; 3) a domain of implementation (hardware, software, or hybrid); 4) performance constraints as mandated by IIoT applications (e.g., RAM, CPU cycles, energy); and 5) specified target deployment (i.e., sensors, gateways, cloud edges). It facilitates the more context-dependent adoption of techniques, which becomes necessary for researchers and practitioners to match the cryptographic options to a range of operational, security, and resource constraints in IIoT. We also incorporate metrics and comparative measurements not found in previous taxonomies, rendering the framework both analytically rigorous and practically applicable.

III. Background

A. Industrial Internet of Things

The IIoT is a big step forward in the integration of advanced technologies. It provides the industry with interconnecting systems that offer higher efficiency, productivity, and innovation. IIoT combines the concepts of the Internet of Things with industrial-grade sensors, devices, and platforms targeted at mission-critical fields, such as manufacturing, energy, and logistics. At the heart of IIoT lies its ability to collect and analyze real-time data, facilitating more intelligent decision-making, predictive maintenance, and automation. IIoT harnesses the power of data-driven insights and machine-to-machine communication to transform the way factories work. In addition, it introduces artificial intelligence (AI), machine learning (ML), and cloud computing to China's industrial ecosystem [12], [13]. Nevertheless, for all its benefits, there are still significant stumbling blocks. These include such challenges as data security, device consistency, and high implementation costs. However, edge computing and the integration of the already mentioned edge way of thinking through blockchain techniques with IIoT may provide new standard solutions to old problems. Research reports the positive impacts of IIoT on industrial processes, from machine-to-machine (M2M) communication and sophisticated data-driven insights that effectively reduce processing times for some important tasks [14], [15]. By adding AI and big data systematic processing, industrial IIoT creates an agile and resilient operational environment. However, the cost of data is enormous and never seems to end. Despite these costs, new technologies, such as edge computing and blockchain are already starting to address the need for efficient use of time and space in IIoT decision-making among heterogeneous devices or across diverse groups (like manufacturers to logistics operators).

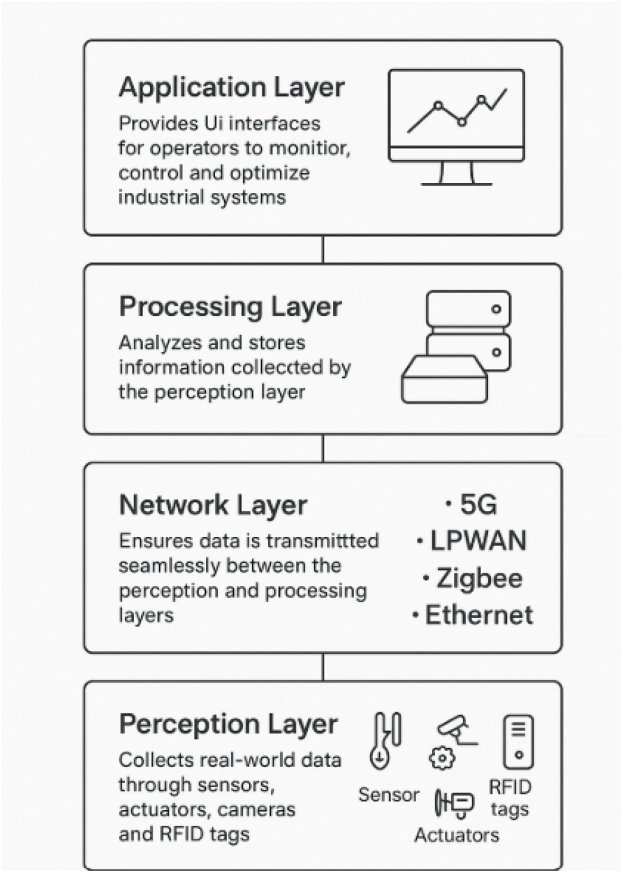

Fig. 1. System model for IIoT.

B. System Model for IIoT

Various layers and components in the system model of the IIoT interact seamlessly to enable data collection, processing, communication, and application under an industrial environment [16], [17]. This multilayered architecture addresses the complexity and diversity of IIoT systems, promising scaling, reliability, and security. Fig. 1 shows the system model for IIoT.

1) The perception layer at the bottom of the IIoT system, collecting real-world data through sensors, actuators, cameras, and RFID tags, and converting physical signals into digital information that makes live industrial process monitoring and control a reality. Temperature sensors, for example, can monitor machinery while actuators adjust operations based on the feedback. Challenges in this layer include ensuring accuracy for all data gathered; scaling up massive deployments, and preventing devices

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from tampering by forces beyond their control. Reliable industrial automation can be achieved only if a sound foundation is provided by this layer.

2) The network layer ensures data is transmitted seamlessly between the perception and processing layers using such technologies as 5G, low power wide area network technologies (LPWAN), Zigbee, and Ethernet. It supports device-to-device communications, which enable decentralized operations and real-time response. The layer must overcome such challenges as maintaining low latency, ensuring data transmission security, and integrating dissimilar communication protocols. 5G, for example, can support high-speed communications between robots and their controllers, while LPWAN is ideal for putting remote sensors into operation. This layer is essential to achieving reliable connectivity and secure data exchange in IIoT systems.
3) The processing layer analyzes and stores information collected by the perception layer, turning raw data into usable insights. Through the use of edge, fog, and cloud computing, this layer provides for localized processing of tasks with low latency and centralized analysis over longer time periods. It also carries out functions like data aggregation, identifying anomalies, and predictive maintenance. Challenges include optimizing workload distributions across different computing layers while maintaining security. Edge computing, for instance, offers immediate response times, while cloud computing can provide advanced analytics. This layer links data collecting with decision making by turning data into actionable information.
4) The application layer provides UI interfaces for operators to monitor, control, and optimize industrial systems. Included in this layer are real-time monitoring dashboards, control of machinery systems (SCADA), and industry logic insight apps. Predictive maintenance software, for example, identifies future machine failures before they occur. Challenges include providing user interfaces that are simple and easy to learn; integrating with older systems should not exclude incorporation into this layer. This layer translates working out less raw data into usable results for operators, so that they may make informed decisions while maintaining their operating efficiency in IIoT environments.

C. Security Challenges in IIoT

At each layer of its extensive interconnectivity and critical role in industrial processes, the IIoT system has a broad range of security challenges [18]. Below, detailed security challenges are presented for each layer, providing insight into the unique threats and vulnerabilities they face.

1) Perception Layer (Edge Devices and Sensors): Sensing, acting, and processing equipment make up the perception layer, which lies between the physical world and the actual IIoT system. Its exposure to the external world makes it an invulnerable point for attack.

a) *Device Authentication:* It is to verify that only authorized devices form the network. Nevertheless, weak authentication protocols can give rise to rogue devices that are inserted into the system by attackers.
b) *Physical Tragedy:* Equipment devices that are deployed in remote or unprotected industrial locations may be manipulated physically, leading to impaired operation or the insertion of malicious hardware.
c) *Eavesdropping:* Wireless data acquisition equipment could be tapped by enemy forces, resulting in the loss of significant information about the industry.
d) *Malicious Firmware:* Attackers may exploit the firmware of edge devices by injecting malicious updates that cause the entire system to fall apart.
e) *Resource Limitations:* The limited computational and energy resources of perception devices have made it difficult to implement robust software security mechanisms, such as encryption without sacrificing performance.

For the perception layer, effective measures like hardware that is tamper-proof, light-weight cryptography, and rigorous device authentication protocols are necessary.

2) Network Layer: Transmission between the perception layer and the processing layer also depends on the network layer. Relying on communication protocols, whether wired or wireless, the network layer is open to interception and modification. For example, because equipment in manufacturing facilities typically attaches to other systems, you cannot trust 100% that data here has not been intercepted in transit.

a) *Data Interception:* By eavesdropping while data is on the move, an attacker can breach sensitive data in sensitive scenarios.
b) *Man-in-the-Middle (MITM) Attacks:* By being in the middle and altering, disrupting, or falsifying communication between devices and central systems, an adversary can disrupt your life.
c) *DoS Attacks:* Busy attackers can pump data at speeds higher than the network can handle, preventing it from doing anything meaningful at all in factories.
d) *Routing Attacks:* Malicious nodes can change data paths by sending them down the wrong channels (misrouting the data), which means packets arrive or are dropped altogether. Offenders will then tend toward having some capability for nonrepudiation of "dependent" things due to their inability to ensure a successful attack without traceable evidence.
e) *Protocol Vulnerabilities:* Many communication protocols, such as MQTT or CoAP, lack built-in security protection measures and are ripe for exploitation.

To counter these problems, end-to-end encryption, secure protocol design, and intrusion detection systems (IDS) are essential for a secure network layer.

3) Processing Layer: As a central part of IIoT systems, it becomes an attractive target for attackers.

a) *Data Leakage:* Industrial sensitive data stored or processed on this level is always subject to unauthorized access or theft by malicious parties.

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b) *Attack by Malware:* Servers at the edge or in the cloud can be attacked by malware, which corrupts data or prevents accurate information.
c) *Unauthorized Access:* With flaws in authentication and access controls available, intruders may infiltrate a system and acquire valuable resources.
d) *Side Channel Attacks:* Attackers can abuse patterns in computational processes to infer sensitive information. This is, such as cryptographic keys, for example.
e) *Insider Threats:* Staff or administrators with ill intent may abuse their access privileges to leak or corrupt confidential data.
f) *Trade-off Between Latency and Security:* In real-time, low-latency data processing must be maintained while strong protection is also required. This can be difficult, however, especially with real-time industrial applications.

To protect this layer against evolving threats, it needs robust access control mechanisms, real-time monitoring, and cutting-edge threat detection systems.

4) Application Layer: The interface for users to interact with the IIoT system is provided by the application layer. Adjoining its output stage is an analytics stage that turns raw data into action-oriented insights-controllable and monetizable. It is ADD vulnerable because of the interface with end users.

a) *Unauthorized User Access:* Weak password policies or misconfigured permissions might mean unauthorized individuals are able to gain access to the system's application layer.
b) *Data Integrity Issues:* Operational decisions derived from falsified or tampered data may result in very expensive errors indeed for industrial operations.
c) *Phishing and Social Engineering:* Attackers may use deceitful methods to elicit sensitive information, such as login credentials, from authorized users.
d) *API Vulnerabilities:* Poorly secured APIs that connect IIoT systems with enterprise applications (e.g. ERP or SCADA) might open the system up to unauthorized access or data leakage.
e) *Out-of-Date Software:* Applications without timely updates or patches are at risk of being exploited by known vulnerabilities, which leaves the system unprotected.

The application layer will require a security policy largely in keeping with the measures in place at corporate networks. Tightening up user authentication, using role-based access control, and regular security audits are essential for security purposes.

D. Overview of Lightweight Cryptography

LWC is a specialized branch of cryptographic techniques tailored to best meet the IIoT device constraints while guaranteeing strong data security, integrity, and authenticity in a generalized manner [19]. It is to address the unique resource-poor environments in which structural integrity and authenticity requirements often come up all at once.

The ability to minimize computational overhead, energy consumption, and memory requirements distinguishes lightweight cryptographic algorithms. These algorithms are specifically optimized for IIoT devices, trying to strike a balance between the strength of security and performance [20]. Concentrating on these points makes it possible for LWC to be used effectively, even on devices with restricted resources.

Its smaller resource requirements mean that it has reduced overhead compared to traditional cryptography methods: caches are emptied, and memory and processing power are needed much less. In addition, LWC's efficiency makes it suitable for use from small-scale end devices all the way up to huge industrial networks [21]. Moreover, lightweight cryptographic techniques strengthen defense against typical threats, such as unauthorized access, data tampering, and replay attacks, heightening IIoT security.

With the help of LWC, data can be kept secret by securing messages between devices. It further aids in the authentication procedures, ensuring that devices and users are true to their identities so that unauthorized access does not occur. Moreover, it makes sure that data remains unchanged and unauthorized–guaranteeing a reliable basis for IIoT systems.

E. Postquantum Cryptography for IIoT

PQC is are cryptographic algorithm that is meant to withstand attacks that can be conducted by quantum computers. Its algorithms do not rely on traditional public-key cryptography algorithms, such as RSA or elliptic curve cryptography (ECC), whose security will be shattered by quantum algorithms like Shor's, but instead on PQC that is based on hard mathematical problems from lattice, code, and hash-based constructions [22].

In IIoT networks, PQC becomes even more crucial since many industrial cyber–physical systems have long durations of use and are mission-critical. Implementations carried out now years are likely to still work ten or more years from now, and hence, we cannot rely solely on classical cryptography for these devices [23].

However, the integration of PQC with IIoT is technically challenging. The key size and memory overhead of most PQC algorithms are much higher than conventional algorithms and can overwhelm the constrained availability of resources on edge nodes and embedded devices. For instance, constructions, such as Kyber or Dilithium, which were recently standardized by NIST, require a few KBs of RAM and storage, which is more appropriate for gateways or high-end IIoT controllers and not really for ultraconstrained sensor nodes.

To meet the security and resource efficiency, hybrid models are being used. In those models, PQC is utilized during an initial handshake or key-exchange phase, where lightweight symmetric tools (e.g., Ascon or AES-GCM) are the workhorses of ongoing communication. This achieves a high quantum resistance while avoiding excessive utilization of the device's computing budget.

PQC is therefore indispensable in the IIoT environment to future-proof the infrastructures, and if wisely chosen, guarantees the long-term security of data confidentiality and

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the sustainability of operations in the quantum-era industrial networks.

F. Energy-Saving Hybrid Cryptography

Energy-saving hybrid cryptography is the smart combination of multiple cryptographic algorithms (often lightweight symmetric algorithms with asymmetric or post-quantum ones) and compromises between security level, power consumption, and computational cost, which is indispensable for IIoT [24].

IIoT devices are deployed in extremely constrained environments, including distant sensors acting in a field, actuators powered by batteries, and passive RFID tags. These nodes cannot apply high-cost encryption algorithms constantly. Hybrid schemes alleviate this limitation by using expensive cryptographic operations (e.g., key exchange, digital signatures) only once during the session establishment and protecting the session using fast symmetric cipher suites (e.g., AES-GCM, Ascon, TinyJAMBU [25], [26]).

For example, secure session key bootstrapping can be realized using an ECC or a NIST-standard post-quantum method, such as Kyber. All subsequent data protection mechanisms use low-energy-consuming block ciphers to ensure confidentiality and integrity. This separation allows us to provide strong security guarantees under very tight device limitations.

Also, some papers study adaptive power-aware encryption, dynamic voltage scaling, and battery-aware cryptographic protocols. They are especially well-suited for energy-scavenging environments in which device recharging is rare or impractical.

Considering the development of future innovations, current research is investigating.

1) AI algorithms for selecting ciphers with real-time battery profiles,
2) Physical-layer and network-layer hybrid frameworks that integrate physical-layer and network-layer encryption,
3) Formal verification techniques to verify adaptive hybrid cryptographic systems.

To sum up, energy-saving hybrid cryptoschemes provide a viable security design for IIoT, to sustain the security of data in the long run, and to improve performance and energy consumption.

G. Motivation for Taxonomy

While lightweight cryptographic procedures demonstrate increasing efficiency, there are no unified classification criteria for such methods. This hampers their integration into industrial IoT systems. Current research is fragmented in several ways. This was true long before any clear framework existed for the organization and comparison of all the techniques. Such fragmentation makes it difficult for practitioners to choose cryptographic methods that are suited to their needs. A systematic classification will help to bridge these gaps by the following.

1) *Organizing Skills:* Categorizing cryptographic methods by such factors as their algorithm type, goals, and efficiency.
2) *Facilitating Comparisons:* Supplying a structure that can be used to weigh the strengths and weaknesses of different approaches.
3) *Guidance for Implementation:* Assisting developers and researchers in selecting the most suitable methods for specific IIoT uses.

This study aims to create a thorough classification of lightweight cryptographic frameworks for IIoT. The Taxonomy is divided into the Algorithm Type of the Technique, the Targets it is trying to achieve, the Venue where it will be applied, the Context of its application, and Energy Efficiency. In addition to being a useful tool for both researchers and practitioners as an indication of current poor research and giving hints on future work, it can also be used as a basis for further study.

IV. Taxonomy-Based Lightweight Cryptographic Frameworks for Secure IIoT

For this study, the classification system is a highly structured outline to show how the IIoT should do encryption, as shown in Fig. 2. Based on cryptographic engineering, it now has a full range of cryptographic items, whether or not they run any one particular kind of algorithm design. Either taking the form of traditional libraries or table-based implementations, new training methods can now be categorized according to the application domain. The Taxonomy method maybe by splitting cryptographic solutions into these kits, the Taxonomy not only simplifies understanding but also pinpoints areas for later research and development.

A. Algorithm Type

This category is about building out algorithms that live at the heart of secure communication schemes within IIoT systems and making computer security software stacks for particular uses we are to ensure that communication is secure, we must also do it efficiently and lightweight cryptographic algorithms are necessary because devices run from limited power sources with next-to-no hope of getting more. In symmetric cryptography, asymmetric cryptography, and hash functions, you may understand this above all else, algorithms reflect the way they work as well as different applications for securing IIoT networks.

1) Symmetric Cryptography:

a) *Lightweight Block Ciphers:* Lightweight block ciphers, such as PRESENT and Speck enable fast, energy-efficient encryption with short key and block lengths. They are a perfect option for deeply constrained IIoT devices, but might have insufficient authentication or not be so guarded against some ambush attacks. Hen et al. [27] focuses on the security challenges of IIoT, which was cobbled together from Rafique et al. [28] 's multifactor authentication protocol. They identify weaknesses, such as the ineffectiveness against insider and KSSTI attacks and the absence of forward security. They argue for a stronger security

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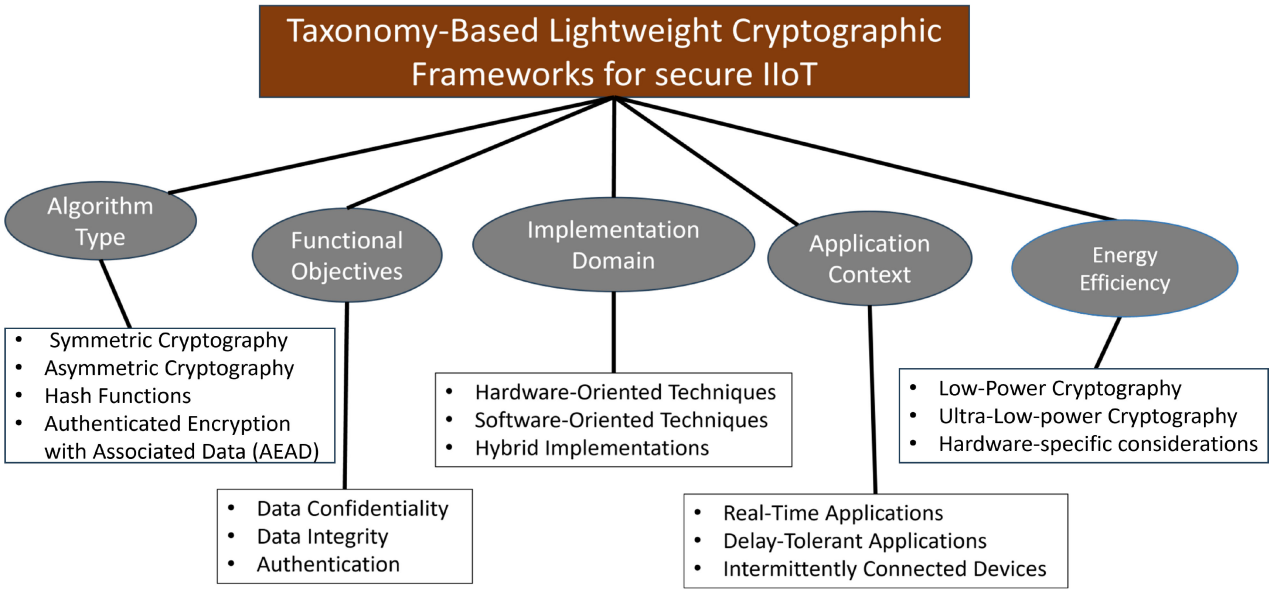

Fig. 2. Taxonomy-based lightweight cryptographic frameworks for secure IIoT.

level and a lower computational cost of their enhancement by providing a formal and informal analysis. Baron et al. [29] study lightweight ML-based IDSs for resource-constrained IoT devices. In their study, the authors compare both decision tree-based and neural network models with respect to IoT-23 and TON_IoT datasets for Raspberry Pi and FPGA platforms. Results focused on the fact that, on average, decision trees outperformed neural networks in both performance and uptake SEE S2, and with FPGA acceleration to fill throughput gaps, are able to facilitate real-time distributed IDS deployment.

b) *Stream Ciphers:* Stream ciphers, like Grain-128a, operate in a bit-by-bit fashion as they were designed for streams of continuously arriving data in IIoT sensors. They provide low latency and energy consumption but rely on secure time synchronization and are subject to stream replays. Ramakrishna et al. [30] propose a novel QoS-based IoT security model that integrates lightweight lattice-based ABE and Quantum Key Distribution for improving the security and privacy of IoT events during data routing in IoT networks. It yields up to 24%, 18%, and 14% improvements on communication overhead, system efficiency, and routing efficiency, respectively, ensuring the aforementioned IoT systems are safer, scalable, and power-efficient for smart cities, smart homes, and smart workplaces. Chang and Chung [31] provides a method based on the WG-8 stream encryption mechanism to reinforce lightweight encryption device authentication against increasing cyber threats in smart manufacturing. Tested on Raspberry Pi3 B+, the method enhances IIoT security at a cost of 40% CPU, suggesting a performance vs real-time protection tradeoff. Ghasemi and Babaie [32] proposes a lightweight RFID authentication scheme that is based on stream cipher with nonlinear pseudo-random number generators (PRNGs) with the help of hybrid high-speed LFSR-based generators (e.g., And2LFSR, Geffe). The scheme has been shown to preserve privacy and to be highly resistant to standard threats of RFID systems (e.g., skimming, cloning, DoS).

c) *Hybrid Symmetric Techniques:* Hybrid symmetric designs use features from both block and stream ciphers for higher efficiency and failure resistance. These are designed for IIoT to tradeoff speed and security, but they are built on top of intricate key distribution mechanisms. Tanveer et al. [33] present LEAF-IIoT, an unobtrusive authentication schema for reliable data exchange in IIoT systems. It provides confidentiality, authentication, and data integrity with authenticated encryption. This framework sets up secure channels and session keys and is resilient to attacks. The evaluation concludes with reduced computational cost and communication complexity in comparison with state-of-the-art models, ensuring its efficiency while maintaining robustness and security. Anand and Singh [34] propose a new secure medical data hiding scheme for industrial IoT scenarios, which specifically deals with the secure protection of outsourced medical images. It incorporates NSST and MR-SVD into robust watermark embedding and uses Fisher-Yates permutation for randomness of dual watermark generation. Post-watermark security is enhanced by a hybrid optimization encryption layer.

2) Asymmetric Cryptography:

a) *ECC:* ECC is also considered to be a strong cryptographic method, even with a shorter private key, which is also suitable for the IIoT devices to exchange the keys safely. However, its implementation requires more computational cost compared to symmetric algorithms. Ahakonye et al. [9] Survey Edge Computing Integration of Blockchain and AI for IoT Network Cybersecurity IDS perspective. Through case studies, the study demonstrates the transformative potential of

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distributed, transparent, and incorruptible IDS by leveraging blockchain technology. However, while facing challenges, such as limited resources, this article portrays the significance of blockchain for IoT security and calls for improvements in privacy-preserving and LWC. Fadia and Toufik [35] presents the cryptographic performance comparison of RSA and ECC in a constrained IoT environment. It almost always comes out on top when compared with ECC in various categories, including memory consumption, energy usage, size of keys, and speed of the process, which makes it perfect for light IoT implementations.

b) *PQC:* Post-quantum algorithms are used for quantum computing-proof encryption. Although secure, existing methods have high memory usage, making their implementation in resource-constrained IIoT difficult. Bagchi et al. [36] present a post-quantum lattice-based aggregate signature scheme under Ring-LWE assumptions for the public blockchain systems, and specifically for Internet of Drones (IoD). It provides both authentication and integrity under quantum-secure assumptions, falls within our class/key application mapping/public-key algorithm and PQC, and is application-specific for aerial IIoT with moderate-to-high performance requirements, thanks to batch verification. Tripathi and Umarao [37] IFDT-BCF-SAS-IIoT, melds Blockchain with ML for tighter security and precision in the IIoT. Through a Fuzzy Decision Tree and a Lightweight Homomorphic Cryptographic Algorithm, this addresses both lightweight operations and transaction management. With Delegated Proof of Work, it can achieve better accuracy, precision, and recall than current methods, verified by its performance, such as RMSE or specificity during both Python-based running and execution. Bagchi et al. [38] present a lattice-based multiauthority ciphertext-policy attribute-based encryption (CP-ABE) scheme that achieves post-quantum security and fine-grained access control. It is especially relevant for a federated IIoT network or multitenant industrial system. It is also in accordance with the post-quantum public-key encryption type, whose concerns are access policy and multidomain authority.

c) *Hybrid Asymmetric Methods:* Tabassum and Rahaman [27] propose a high-end multifactor authentication (MFA) Protocol and Agreement on IIoT, in response to the increasingly interconnected world. This not only has dynamic authentication on different levels of risks and advanced encryption, but it also involves efficient, secure key exchange methods. In a likely operation, it offers expansion and adaptation so as to keep up functioning levels along with more protection against agile, advanced cyber threats. Yogaraja et al. [39] describe an energy-efficient secure encryption model for IoT applications using ECC and the adaptive hunter prey optimization (AHPO) method that guarantees efficient key generation. ECC provides a light and strong cipher, while AHPO effectively improves the key management capability on these resource-limited networks.

3) Hash Functions:

a) *Lightweight Hashing Algorithms:* These techniques use a digest, such as SHA-256 for integrity and authentication without the use of encryption. They are also lightweight and efficient but are not ideal if you need privacy. Within IIoT environments, Paliwal [40] raise the issue of security challenges in wireless sensor networks (WSNs). They accordingly propose a hash-based conditional privacy-preserving authentication and probabilistic key exchange protocol, which is lightweight in nature. In contrast to existing works, they point out vulnerabilities in Xiong et al.'s ECC-based method. By way of simulations in AVISPA and Real-or-Random oracle models, the protocol demonstrates its robustness against known attacks. Thus, it allows efficient and secure communication with no more than a minimal computational overhead. Zhang et al. [41] proposed an effective authentication and key agreement (AKA) protocol for IIoT by taking into account two major deficiencies in recent methods, including the vulnerability to attacks and the huge computational overhead. Utilizing chameleon hash functions with PUFs, the design achieves robust protection against impersonation and device capture and keeps lightweight for resource-constrained IIoT devices.

b) *Hash-Based Authentication:* Bouayad et al. [42] came up with the idea of "Lightweight-Fed-network IDS (NIDS)." It uses federated learning (FL) as its core principle in order to enhance the efficiency and scalability of NIDS. The method applies structured model pruning; training times are sped up by up to three times, and model size is reduced by more than 90%. Detecting anomaly events occurs at a very high rate (99%), while resource-efficient privacy is guaranteed, and it is thus well suited for practical application in today s NIDS. Tandel and Nasriwala [43] contribute to post-quantum security in the context of Industry 4.0 and present a hash-based signature scheme that is suitable for secure authentication in the IoT with lightweight devices. Because the quantum attack on public-key cryptography reduces security, this work researches collision-resistant hash functions and introduces Blake2s as lightweight, high-performance alternative to SHA-256. Experimental results indicate that the key generation computation of the signature scheme can be sped up by 27.18% of the original version and save memory.

c) *Hash Chains:* Bhaskaran et al. [44] present BC-LWCIE technology for secure image encryption in Industry 4.0/5.0 settings, a blockchain-enabled cryptographic framework. Aiming at maximal PSNR for image quality while using light hash functions and optimal key generation by the chicken swarm optimization algorithm. The data that is put onto the public chain is encrypted, which adds another layer of secrecy for IIoT. This work by Bera et al. [45] proposes a private blockchain, AI, and fog computing-based based to protect data from wearables IoMT devices. It employs lightweight symmetric encryption and blockchain consensus to maintain

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confidentiality and integrity in a resource-constrained environment in the healthcare community.

4) Authenticated Encryption With Associated Data: Authenticated encryption with associated data (AEAD) algorithms, such as Ascon and AES-GCM provide both confidentiality and integrity of the data. They are also well-suited for secure messaging in industrial IoT, where metadata needs to be available but tamper-evidently protected. De Santis et al. [46] aimed to provide efficient, constant-time software implementations of ChaCha20, Poly1305, and their AEAD combination for ARM Cortex-M4 processors, targeting secure, lightweight TLS communication in IoT. Cagua et al. [47] assess the NIST-approved lightweight encryption algorithm, ASCON, for use in secure IoT communication. We show that ASCON is low latency and computationally efficient, and in combination with experiments on Raspberry Pi, we confirm that it is suitable for real-world IoT in constrained environments that necessitate securing AEAD with a lightweight scheme. Vidhya [48] proposed an AES-GCM-based FPE scheme specially for structured cloud data. It remained faithful to the original data formats and types and solved the schema integrity problem in encrypted databases. AES-GCM provided the guarantees of confidentiality, integrity, and replay resistance and was therefore very suitable for secure and efficient cloud-based data protection. Rahman et al. [49] study the key-commitment security of cipher-based AEADs, such as Eevee, PAEF, and SAEF. Although these schemes are efficient for IoT and MPC, they may not be resilient against ciphertext collisions under different keys. This article cautions against using such modes in applications requiring strong key-commitment guarantees.

5) Discussion: As listed in Table II, the type of algorithm you use affects the tradeoff between security strength and computational efficiency. Symmetric encryption has high efficiency, which is well-suited for low-resource-constrained devices, while asymmetric cryptography, like ECC, can guarantee strong security. Hash functions are lightweight and efficient by their very nature.

TABLE II
COMPARISON OF ALGORITHM TYPES

Algorithm Type	Research Study	Strengths	Limitations
Lightweight Block Ciphers	Han et al. [27]	High efficiency, low resource use	Vulnerable to insider attacks if not implemented robustly
Stream Ciphers	Ramakrishna et al. [30]	Scalable, quantum-resilient, and power-efficient	Computational complexity in dynamic QoS-based IoT environments
Hybrid Symmetric	Tanveer et al. [33]	Resilient to attacks, low computational cost	Dependency on secure channel establishment
ECC	Fadia et al. [35]	High security with minimal key size	Requires additional optimization for resource-constrained devices
Post-Quantum	Tripathi et al. [37]	Strong against quantum attacks	Complexity in implementation for large-scale IIoT systems
Hash-Based Methods	Paliwal et al. [40], Bhaskaran et al. [44]	Efficient, minimal computational overhead	Limited scalability for high-throughput IIoT systems
AEAD	Cagua et al. [49], Rahman et al. [49]	High integrity and confidentiality, low latency (Ascon), lightweight deployment feasible on constrained devices	Vulnerable to key-commitment attacks (e.g., Eevee, SAEF), needs careful design in for cipher-based schemes

B. Functional Objectives

In IIoT systems, cryptographic techniques are often adapted to meet individual security objectives. Data confidentiality, data integrity, and authentication are the three main objectives of this section. These are the basic requirements for any secure communication system, ensuring protection to make up (as well as trust and accuracy) of data that is transferred In IIoT environments.

1) Data Confidentiality:

a) *Lightweight Encryption:* In the Vehicular Fog System, Kong et al. [50] present a lightweight blockchain-based authenticated key management system for Cloud-Assisted IIoT. Using elliptic curves and hash algorithms, the system provides facilities for lightweight client authentication, dynamic management of client connections, and guaranteed freshness of cryptographic keys. With results obtained through empirical verification and Scyther tool analysis, it shows security and efficiency characteristics. Eliminating single-point failure, while suggesting topics for future research, the method is empirically validated.

b) *Tokenization:* Reddy and Koti [51] propose a blockchain-based authenticated key management system for Cloud-Assisted IIoT to overcome challenges like resource constraints and physical access vulnerabilities. Using elliptic curves and hash algorithms, the scheme supports lightweight authentication, dynamic client management, and key freshness. Empirical analysis via the Scyther tool confirms its security and efficiency, addressing single-point failure while suggesting future research directions.

c) *Confidentiality in Networks:* Yu et al. [52] present a lightweight blockchain-based authenticated key management system for Cloud-Assisted IIoT. Using elliptic curves and hash algorithms, the system provides facilities for lightweight client authentication, dynamic management of client connections, and guaranteed freshness of cryptographic keys. With results obtained through empirical verification and Scyther tool analysis, it shows security and efficiency characteristics. Eliminating single-point failure, while suggesting topics for future research, the method is empirically validated.

2) Data Integrity:

a) *Hash-Based Integrity Checks:* Introducing a security interaction protocol which is designed for the needs of the energy Internet, Chen et al. [53] point out problems, such as transmission of data in plain text, changes in the data by eavesdropping manipulation, or no confirmation at all. The protocol is able to guarantee data privacy, data nontampering, and fine-grained access control through additive homomorphic encryption,

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multilevel hash algorithms, and attribute-based access control.

b) *Redundancy Techniques:* Lavanya et al. [54] propose a secure cluster-based data aggregation protocol (SCDAP) in order to improve WSNs security and reduce energy consumption. The protocol—by itself not already a network—takes responsibility for establishing secure clusters that are in charge of data aggregation, effectively negating redundant information that arises therefrom. It also implements robust authentication and key management to ensure the security of data traffic on these clusters.

c) *Multilevel Integrity:* Logrippo [55] discusses methods of data flow control for security considerations during the Internet of Things and Cloud service environments. This article sets out a basic way to mechanize partial orders of equivalence classes and proposes simplification as well as generalization for traditional lattice-based theories. Labeling systems map objects onto these orders. To assure the security of vital algorithms, such as efficient transitive closure and fast recognition on strongly connected components, algorithms are incorporated into the system for performing labels.

3) Authentication:

a) *Key-Based Authentication:* Megala et al. [56] address the security issues in connecting blockchain with IoT, such as eavesdropping, data outcross illumination, and terminal failure, under the belief that through the use of an advanced public key password scheme to sign and authenticate without using extra programs, people will obtain device- and data integrity.

b) *Multifactor Authentication:* Aljanah et al. [57] employ traditional public-key cryptographic algorithms for secure authentication and key management; it also offers fault-tolerant capabilities. With data integrity being assured by the basic security protocols of key management alone, no single system can fail (and of course, all the data in each system). The analysis shows that blockchain has the potential to promote IoT security and its effect on traditional distributed cloud systems.

c) *Decentralized Authentication:* Shahidinejad and Abawajy [10] develop their M2I architecture to fit into the authentication and authorization schema of an IoT cluster, obtaining systems similar not completely identical, to what exists in a data center. They propose two protocols—hybrid group authentication and key acquisition (HGAKA) and hybrid group access (HGA)–working at different layers of IoT technology, but both utilize symmetric and asymmetric cryptography for multifactor group authentication.

4) Discussion: As listed in Table III, Cryptographic methods pursue specific security goals in IIoT. Lightweight encryption ensures safe communication channels, redundancy techniques enhance reliability, and multilevel authentication ensures trust and robustness.

TABLE III
COMPARISON OF FUNCTIONAL OBJECTIVES

Objective	Research Study	Strengths	Limitations
Lightweight Encryption	Kong et al. [50]	Eliminates single-point failure	Scalability challenges in large-scale deployments
Hash-Based Integrity	Chen et al. [53]	Guarantees privacy and tamper-resistance	Limited efficiency in dynamic IIoT environments
Redundancy Techniques	Lavanya et al. [54]	Improves energy efficiency and data aggregation	Requires robust key management for secure clusters
Multi-Level Authentication	Shahidinejad et al. [10]	Provides flexible and layered security	computational cost for frequent multifactor verification

C. Implementation Domain

The implementation domain decides whether cryptographic techniques are hardware-based or software-based, or a mixture of the two. Hardware-based techniques carry out real-time processing quickly and efficiently, while software-based approaches allow greater flexibility of the process to occur on the fly. In IIoT systems hybrid implementation combines strengths from both sides. This gives an appropriate level of performance and shows an ability to scale with demand.

1) Hardware-Oriented Techniques:

a) *ASIC Implementation:* Singh et al. [58] proposed two lightweight hardware realizations of LBlock in order to apply the cipher toward digital imagery protection, particularly for high-security demands such as military and biometric functions. Both design procedures are applicable to FPGA circuits. Under the 1 Xilinx Virtex-5 FPGA VCU330S performance space, they make use of only 56 slices at 492.441 MHz to fulfill their tasks covertly and automatically without human input.

b) *FPGA-Based Designs:* Teodoro et al. [59] propose that the tree parity machine (TPM), a neural network-based cryptographic technique, be used for rapid key exchange in resource-scarce IoT system devices. Unlike traditional methods of security, TPM skips number theory and thus reduces computational expense. FPGA implementations were optimized with respect to response time, operational frequency, and resource consumption, outperforming multicore CPU program instances.

c) *Energy-Efficient Sensors:* Raza et al. [60] presents a lightweight AKA scheme designed specifically for use in SDIoT environments in smart homes. The scheme incorporates registration, authentication, and the selection of key phases with class-specific access control. Security validation using the AVISPA tool indicates that it is robust against attacks. The simulation results show that it is not only feasible but also outperforms traditional systems in terms of operational time, computation hardness, and energy efficiency.

2) Software-Oriented Techniques:

a) *Resource-Constrained Protocols:* Varo [61] observed in their study the burgeoning "dual challenge" of cybersecurity combined with energy efficiency in IoT environments, especially so for devices that must survive

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on only small amounts of electricity from batteries. What they propose is a dynamic mechanism that uses reduced round cryptography that scales back security a bit, but also depends on how much energy is left in the device's batteries.

b) *Low-Latency Algorithms:* Kuznetsov et al. [62] carry out a complete analysis of high-speed stream ciphers that are necessary for providing security in wireless networks. It first examines the design principles and challenges of high throughput combined with complicated mobility networks. A Method is proposed here for evaluating cipher efficiency and security in different environments via model emulation of this process in simulation systems.

c) *Cross-Layer Implementations:* Resner et al. [63] indicates a new stage in the evolution of encryption technology. Their trustful space–time protocol (TSTP) brings together georeferencing, encapsulation, encryption, integrity verification, and data storage in a cross-layer, message-based sequence protocol. TSTP eliminates the need for monolithic network services which is too common in today's protocol design by using component-based, event-driven architecture.

3) Hybrid Implementations:

a) *Hardware-Software Co-Design:* Pereira [64] discuss how embedded systems have evolved to become more and more sophisticated and diverse from the days of single-use, resource-limited devices. Modern embedded applications make use of general-purpose operating systems (GPOSs) for interface and computation tasks. However, because they have strict timing requirements attached to them, this calls for everything to be done in coordination with temporal goals. Bhatta et al. [65] give a comprehensive analysis of SRAM-PUFs, based on its architecture, principles for design, and application into hardware security. They evaluate its performance characteristics, learn about weaknesses found, and suggest approaches for improvement through real-world integration.

b) *Adaptive Protocols:* Cadenelli [66] propose a new method to tackle massive information requirements through hardware and software co-design, aimed at reducing the high costs of data-intensive genomics applications. This research aims to reduce the total cost of ownership for genomics workloads by optimizing with algorithms, using nonvolatile memory (NVMs) as well as leveraging powerful GPUs and FPGAs for acceleration.

c) *Real-Time Verification:* Nguyen and Kortun [67] focus on questions of real-time optimization in IIoT, where data flow across the network is now tremendous and environments are dynamic. They place the greatest emphasis on the importance of embedded real-time optimization (of type following graves), able to provide rapid updates so that signal processing deadlines are met rigidly in time without missing a beat.

4) Discussion: As listed in Table IV, the implementation domain is a tradeoff between flexibility and performance. Hardware-oriented techniques are fast but inflexible; software-based methods permit hybrid implementation.

TABLE IV
COMPARISON OF IMPLEMENTATION DOMAINS

Domain	Research Study	Strengths	Limitations
Hardware-Oriented	Singh et al. [58], Teodoro et al. [62]	High speed operations, optimized for certain applications	can not be generally adaptable to IIOT context
Software-Oriented	Varo et al. [61]	Dynamic, scalable for several different IoT systems	real-time requirements may not be met
Hybrid Implementations	Pereira et al. [64], Bhatta et al. [65]	Combines speed and flexibility	Increasing implementation badly constraints overall system performance

D. Application Context

This section looks at the different operating environments in which cryptographic techniques are used. The taxonomy divides these scenarios into real-time applications, delay-tolerant applications, and intermittently connected devices, reflecting the diverse, unpredictable nature of IIoT networks. This helps you to select cryptographic techniques that meet both the requirements for latency and reliability It ensures interaction with other devices.

1) Real-Time Applications:

a) *Critical System Monitoring:* Ooi et al. [68] proposed a complete IoT solution that monitors outdated manufacturing machines to track changes in production. They aim to help SMEs get started with Industry 4.0 chip practice. At work, the system has reached a reliable encryption key management mechanism to secure the communication interface and has shown a strong practical impact for 365 days running with no loss of data in a real manufacturing environment.

b) *Time-Sensitive Data Streams*: According to Sisinni and Mahmood [69], LPWAN have become a key component in the IIoT system. This not only improves production efficiency but also allows for more effective machine-to-machine communication. The flexibility, scalability, low-power characteristics, and other benefits of LPWAN are addressed in this chapter, with emphasis on standard LoRaWAN, which has achieved significant attention from academia as well as industry.

c) *Latency-Sensitive Networks:* In their recent review of interoperability problems with IIoT Hazra et al. [11] turned their attention to integrating the various devices, tools, and protocols of Industry 4.0. They analyzed the latest IIoT techniques, models, and solutions for these problems and learned from developments in architecture that were favorably regarded as state-of-the-art.

2) Delay-Tolerant Applications:

a) *Archival Data Security:* This research is done by Rodriguez et. al. [70] that describes how 5G—the mobile communications system and a key technology for industrial automation 4.0 at present–fits together and provides new angles of attack on problems that must be

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solved before 5G can be successfully adopted. The focus (partial emphasis should be given to their approach) is the structured integration concept: its operational flow and tools were inspired by both analytic models and experiments.

b) *Asynchronous Communication:* Sangeetha and Raja [71] investigated IoT's role in intelligent power grids. They list factors like unidirectional data flows and traditional electrical networks that do not comply with energy efficiency directives. They turn to IoT-enabled connectivity, automation, and monitoring for energy generation and distribution. Here, SG scheming and application software systems are briefly touched on, as well as multifaceted topics with IoT machinery digitizing electric grids as SG; energy efficiency combined to run in series.

c) *Backup Systems:* Gillani et al. [72] provided a complete survey of data collection in protocols for vehicular ad hoc networkss (VANETs). This is one of the main aspects of intelligent transport systems (ITS). They categorize protocols as best-effort, delay-tolerant, or real-time (for clarity, they introduce a simple taxonomy). Parameters such as delivery ratio and recovery strategies are the subjects of comparison.

3) Intermittently Connected Devices:

a) *Edge Device Security:* To address the issues of security, privacy, and a centralized infrastructure that exist in IoT applications, Pajooh et al. [73] look at how 6G-enabled IoT technology can be combined with blockchain techniques. The authors explore blockchain tech's potential in the realm of decentralized access control. It also has the potential to get rid of single points of failure and forgo third-party authorities wherever trust is concerned.

b) *Opportunistic Communication:* Sanchez and Cano [74] examine low-power wide area networking (LP-WAN) as a paradigm for addressing the energy efficiency and reliability needs of industrial IIoT networks. They review key LP-WAN solutions, discuss their advantages and limitations, and analyze their deployment status in Spain. Despite being in early development stages, LP-WAN systems show strong potential for advancing IIoT networks and services in industrial applications.

c) *Offline Key Exchange:* Fan et al. [75] introduced the SAKE * protocol. It is a lightweight solution for authentication and key exchange (AKE) oriented toward IIoT environments. SAKE* guarantees that the plaintext and encryption output from previous communications are released, leveraged in this alert state attack (ASA) method key, and become useless to later generations with PFS.

4) Discussion: As listed in Table V, different IIoT applications have their own unique requirements. Real-time applications make low latency a priority at what cost in terms of reliability; delay-tolerant systems focus on reliability alone; intermittently connected devices must handle varying levels of load and connectivity.

TABLE V
COMPARISON OF APPLICATION CONTEXTS

Context	Research Study	Strengths	Limitations
Real-Time Monitoring	Ooi et al. [68], Sisinni et al. [69]	Reliable encryption, low latency	Limited support for large-scale real-time data streams
Delay-Tolerant Applications	Rodriguez et al. [70], Gillani et al. [72]	Ensures data security in archival systems	Performance trade-offs in terms of retrieval speed
Intermittently Connected Devices	Pajooh et al. [73], Fan et al. [75]	Decentralized, eliminates single points of failure	Complexity in managing opportunistic communication

E. Energy Efficiency

Energy efficiency is at the core of any cryptographic techniques designed for IIoT devices, which use many systems. It does not yield immediate results. This category focuses on low-power cryptography and ultralow-power cryptography. It addresses the need for operations in resource-constrained environments to run smoothly and for a long time. Techniques like power-aware key management and energy scavenging promise security, yet keep the duration and functionality of IIoT devices at an acceptable level.

1) Low-Power Cryptography:

a) *Power-Aware Key Management:* Humayun et al. [76] demonstrate the promise of 5G-augmented IIoT to reshape sectors of activity such as smart city development, health care, and manufacturing. They also address the pivotal problems of privacy arising from interoperable devices and energy efficiency. To overcome these lacks, they propose a systematic framework for ensuring both privacy as well as energy saving in 5G-based Industrial 4.0. Rigorous theoretical analysis as well as case studies show that the model provides insights for methods to improve both privacy and sustainability in future 5G systems.

b) *Dynamic Power Scaling:* Farajallah et al. [77] concentrate on the issues of energy and security in WSNs with energy harvesting. In this regard, they propose a Deadline Mechanism to manage fluctuating energy levels dynamically while ensuring data security through adaptive encryption. Furthermore, they also come up with a novel chaos-based cryptosystem, robust against known attacks and very much faster than what has gone before: 7× more so even than AES. This is faster, even in comparison with any other chaos-based system you could name. The proposed method combines low complexity and optimum long-term data security and performance, thus promoting WSN sustainability and efficiency. Haripriya et al. [78] falls within one of the scopes of the green communication area, namely, solving the global power deficiency by proposing a power-saving UART designed on a 28 nm based Artix-7 FPGA. Targeting industrial IoT and communication applications, it uses voltage scaling methods to cut energy consumption.

c) *Battery-Sensitive Algorithms:* Hossain [79] examine methods for establishing lightweight keys, and

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consider one called OSCORE together with another called EDHOC, which improves the security of IoT networks confronted by scarce resources. In testing the performance of these two protocols within a physical system, its results will include how much hardware acceleration is needed, the computational overhead incurred, and power consumption. Krot et al. [80] propose to predict the State of Health (SoH) of a lithium NMC battery in autonomous mobile robots using the discharge feature model. The model allows real-time mission planning and energy prediction, regardless of the type or the duration of the mission, using reliable battery behavior.

2) Ultralow-Power Cryptography:

a) *RFID-Specific Ciphers:* Gong et al. [81] presented an RFID secure, lightweight, rapid authentication protocol that is particularly suitable for traffic congestion cases and IoV. By combining ECC with hash functions and the use of edge servers, this protocol takes care of secure vehicle data management matters, but at the same time ensures low computational load or network overheads. Sivagurunathan and Ganeshan, [82] provides a complete study of LWC algorithms for IIoT and Industry 5.0 systems. Mainly targeted at devices that have constrained memory, power, and processing capabilities, it presents the difficulties of providing authentication, secrecy, and integrity in restricted networks.

b) *Energy Harvesting Devices:* Sanislav et al. [83] look to energy scavenging (EH) as a feasible series of unsustainable environmental protection programs for corrupting ancient, tired battery power IoT connectors, and further reducing network life expectancy. Consider the reviewer breaks EH's ecological (no battery), and economic, and is used to require less maintenance. Case studies of recent developments in EH techniques are given, and the significance of large-scale EH assessment carried out on IoT. Markiewicz et al. [84] propose a new approach to improving TEGs for IoT energy harvesting based on waste heat and mechanical vibrations. The novelty is the use of a variable conductivity medium, which is a fluid with air gaps, from the TEG to the heat sink. The random motion results in splashing, forming temporary thermal bridges to enhance heat transfer and increase the TEG performance. Moloudian et al. [85] examine energy harvesting solutions for battery-less IoT devices in 5G-based, green Industry 4.0 networks. Focusing on RF-based (wireless) harvesting as well as potential new RF methods, this book offers valuable sustainable solutions for the development of WSNs. This article presents new methods like multisource hybrid harvesting and additive manufacturing for personalized energy systems.

c) *Passive Device Encryption:* Wang et al. [86] propose a blockchain-based data-sharing model for the IIoT, which is secure and efficient for proxy re-encryption. The protocol provides for reliable data sources and access control by means of a two-layer authentication scheme for storage and retrieval. To combat the storage restriction of blockchain, we propose an on-chain/off-chain hybrid storage and a data-packing strategy to improve storage efficiency. Galappaththige et al. [87] examine passive IoT powered by backscatter communication (BackCom) as a way of meeting the need for massive connectivity in 5G and beyond. They offer a comprehensive survey, including BackCom designs, tag elements, and channel characteristics. A comprehensive link budget analysis is given, supported by numerical data and simulation results. Wang et al. [88] presents a new physical layer encryption and authentication (PLEA) scheme designed for energy-efficient IIoT devices, combining lightweight encryption and authentication at the physical layer.

3) Hardware-Specific Considerations: Van Leemput et al. [89] introduce a 6TiSCH low-power node (6LPN) that significantly cuts down on energy spent for TSCH joining and operation by up to 94%-via optimized scanning and downlink traffic handling. It can reach low latency and be fully compatible with 6TiSCH, which is well-suited for the low idle current of battery-operated IoT devices. Krishnamoorthy et al. [90] investigated optimized SoC selection for heterogeneous IoT applications using a genetic algorithm. Through Zynq-7000 FPGA-based microprocessor architecture evaluation with performance metrics, it achieves 84.62% accuracy in SoC's silicon type identification, leading IoT hardware design researchers in real-time, cost-effective SoC hardware design. Krebs and Schneider [91] presented virtual channels on top of physical channels to minimize memory usage in the synthesized code of dataflow process networks. Sharing buffer space dynamically, the approach thus achieves up to 50% savings in memory without affecting performance, although it may lead to new sorts of deadlocks because of contention on shared memory.

4) Discussion: As listed in Table VI, energy-efficient cryptographic methods guarantee the sustainability of IIoT devices, with devices that support a limited amount of energy sources using low-power and ultralow-power techniques.

V. Comparison of Performance Parameters

This section examines by performance metrics of lightweight cryptographic techniques in terms of computational efficiency, energy consumption, and memory requirements; meanwhile, it also explains how these techniques adapt to the needs of resource-starved environments.

5) Overview of Performance Metrics for Light Crypto: The evaluation indicators for lightweight cryptographic techniques primarily include three aspects, all of which can be defended mathematically to allow for exactness.

a) *Computational Efficiency:* Computational efficiency C_E is the ratio of the number of operations completed (O) to the time taken (T)

$$C_E = \frac{O}{T}.$$

Here, O stands for the operations required for cryptographic purposes (e.g., encryption, decryption, or squaring), and T is the time taken for these operations.

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TABLE VI
COMPARISON OF ENERGY EFFICIENCY

Efficiency	Research Study	Strengths	Limitations
Power-Aware Management	Humayun et al. [76]	Reduces energy consumption while maintaining security	Limited performance under high-load conditions
Dynamic Power Scaling	Farajallah et al. [77]	Adapts security based on energy levels	May compromise security in extreme low-power scenarios
RFID-Specific Ciphers	Gong et al. [81]	Low computational overhead, suitable for IoV	Limited to RFID-specific scenarios
Energy Harvesting Devices	Sanislav et al. [83]	Promotes sustainability through energy scavenging	Dependency on environmental energy availability
Passive Device Encryption	Galappaththige et al. [87]	Enables operation without active power sources	Scalability challenges for large-scale IIoT systems
Low-Power Protocol Optimization	Van et al. [89]	Achieves up to 94% energy reduction via optimized 6TiSCH node design	Limited to TSCH-based protocols
Hardware-Level Optimization	Krishnamoorthy et al. [90], Krebs et al. [91]	SoC selection via GA and dynamic memory sharing yield high accuracy and up to 50% memory savings	Risk of memory contention-induced deadlocks and platform-specific constraints

A higher value indicates better performance of a given algorithm–quite critical in IIoT systems affected by such delays.

b) *Energy Consumption:*
The energy cost E_C of a cryptographic operation can be modeled as

$$E_C = P \cdot T$$

where P is power consumption during the operation (in watts), and T the time required (in seconds). Devices with limited energy need to keep E_C as low as possible if their lifespan is to be prolonged.

c) *Memory Requirements:* Memory requirements M_R depend upon the aggregate of storage space needed for keys (K), algorithms (A), and operational states (S)

$$M_R = K + A + S.$$

Here, K is the size of the cryptographic key, A the memory footprint of the algorithm, and S the additional states (e.g., initial values or nonces) needed during execution
Efficient cryptographic techniques reduce M_R while maintaining the same level of security.

A. Comparison Mercies

From the Table VII, we can see

1) *Computational Efficiency:* Performance-wise, techniques such as lightweight block ciphers and Hash Functions are supreme, showing potential fitting use in high-latency applications. Multilevel Authentication and ECC, while secure, exhibit a certain overhead loss.

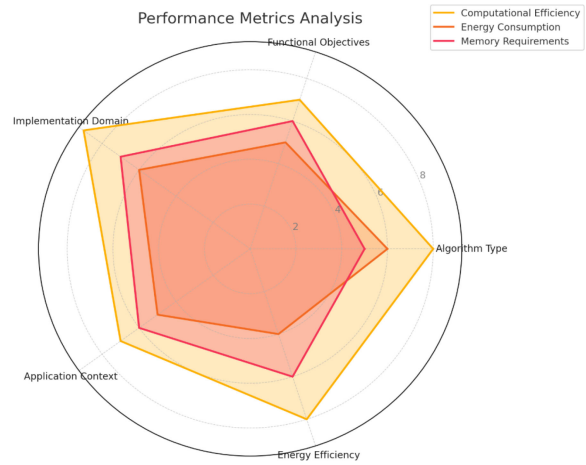

Fig. 3. Performance metrics analysis.

2) *Energy Consumption:* Low energy consumption is one consideration, and in this respect, RFID-specific ciphers and power-aware management hold great promise for low-power devices. Techniques involving a multistep method, such as Multilevel Authentication, are less efficient in terms of energy consumption.
3) *Memory Requirements:* Hash functions and lightweight encryption require little memory, but ECC and those techniques used on intermittently connected devices may need more memory due to the size of their keys and operational states.

B. Implications for IIoT

The conclusions highlight the importance of selecting cryptographic techniques for IIoT applications according to the particular requirements and restraints of those environments. For example:

1) Real-time applications benefit directly from low-computation+low-power devices.
2) Delay-tolerant applications need to use power-efficient techniques to maximize the life of the device.
3) Intermittently connected devices must find a compromise between energy consumption and memory usage to function continuously.

C. Visual Analysis

Performance metrics illustrated by the radar graph in Fig. 3, demonstrate that Algorithm Type and Implementation Domain exhibit strong computational efficiency to a degree suited for real-time or latency-sensitive applications. Elements/stages in these divisions, such as hardware-oriented encryption or lightweight block ciphers, optimize for quick speeds while keeping memory requirements within reason and power consumption not more than moderate. On the other hand, Functional Objectives and Application Context achieve good performance on all parameters, the slightly higher energy consumption as well as memory requirements reflect added complexity to meet objectives such as multilevel authentication or support of intermittent connected devices.

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TABLE VII
QUANTITATIVE COMPARISON OF PERFORMANCE METRICS ACROSS CATEGORIES

Category	Techniques	Computational Efficiency (Clock Cycles)	Energy Consumption (mJ/bit)	Memory Requirements (ROM/RAM KB)
Algorithm Type	Lightweight Block Ciphers (e.g., PRESENT, Speck)	9,000–11,000	0.060–0.090	2.0 / 0.5
	ECC (e.g., ECC-P192)	70,000–90,000	0.350–0.500	16.0 / 2.5
	Hash Functions (e.g., SHA-256, PHOTON)	12,000–18,000	0.080–0.200	4.5–10.5 / 1.0–1.6
Functional Objectives	Lightweight Encryption (e.g., Ascon, TinyJAMBU)	8,000–12,500	0.042–0.065	2.3–3.2 / 0.6–0.9
	Redundancy Techniques (e.g., hash + MAC)	15,000–20,000	0.100–0.180	5.0 / 1.0
	Multi-Level Authentication (e.g., ECC + Hash)	>80,000	0.300–0.500	16.5 / 3.0
Implementation Domain	Hardware-Oriented (ASIC/FPGA)	5,000–9,000	0.030–0.070	Depends on embedded core
	Software-Oriented (MCU-based)	10,000–25,000	0.100–0.250	5.0–12.0 / 1.0–2.0
	Hybrid Implementations	9,000–15,000	0.070–0.150	6.0 / 1.5
Application Context	Real-Time Monitoring (e.g., AES-GCM, Ascon)	< 15,000	0.065–0.120	5.5 / 1.0
	Delay-tolerant Applications	15,000–30,000	0.120–0.200	8.0 / 2.0
	Intermittently Connected Devices	>25,000	0.180–0.300	10.0–16.0 / 2.5–4.0
Energy Efficiency	Power-Aware Management	Adaptive (Varies by duty cycle)	<0.050 (optimized)	Varies by stack
	RFID-Specific Ciphers (e.g., Grain, LBlock)	7,500–10,000	<0.050	2.0 / 0.5
	Energy Harvesting Devices	> 15,000	<0.100	12.0 / 2.0

Fig. 4. Performance metrics comparison across categories.

Category Energy Efficiency should not be overlooked, as seen in Fig. 4, it has very low energy consumption. Strategies to achieve this usage of lesser power include energy harvesting and cryptography tailored for RFID, so now it is a necessity for components working under strict power constraints. But this efficiency comes at moderate memory costs: for many of these methods, the protocols involved are specialized. This view is underscored by the bar chart in Fig. 4, where it can be seen that energy efficiency with the smallest consumption through the radar domain and algorithm type, computed on a different scale, is minimizing energy wastage most consistently across all categories.

Thus, real-time and latency-sensitive systems should depend on whether the time to how fast computationally efficient methods from Algorithm Type or Implementation Domain are a priority. On the other hand, devices required to operate with minimal maintenance for days at an end should employ energy-efficient methods; likewise those systems with intermittent connection but in need nonetheless of frequent operation Application Context techniques might solve such taints throughout its whole structure The competitive analysis highlights the importance of aligning encryption methods with the unique characteristics of IIoT environment if we are to realize both safety and efficiency.

VI. RESEARCH GAPS AND FUTURE DIRECTIONS

A. Research Gaps

The analysis of LWC techniques for IIoT exported many of them and polished and improved others so that even a great noncryptologist could read them over leisurely.

1) *Integration of Postquantum Cryptology is Limited:* PQC has been studied in papers like Tripathi and Umrao [37]. But how to actually use this practical information on resource-poor IIoT systems remains unexplored. The nonexistence of real-life prototypes and the absence of performance data in its discrete steps all make us suspicious of whether it is feasible at all.
2) *Scalability in Large-Scale IIoT Authentication:* Authentication protocols, such as multifactor and key-based approaches, usually suffer from scalability problems in large and heterogeneous IIoT networks. Shahidinejad and Abawajy [10] argue that high

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computational and communication overhead is due to higher authentication load and node membership dynamics. However, promising scalable mechanisms, e.g., hierarchical key management or group-based authentication, have not been extensively investigated.

3) *Energy-efficient Hybrid Implementation:* Hybrid approaches (e.g., Bhatta et al. [65]) take advantage of both software and hardware, but little is known about their energy requirements when applied under dynamic operating conditions such as fluctuating workloads or in an environment where all this simply cannot be controlled.
4) *Interoperation and Standardization:* The reviewed studies appear to have a different technology for each case studied, yet this diversity across IIoT platforms presents huge problems. For example, Hazra et al. [11] call for standardization in order to make different systems of heterogeneous nature able to communicate seamlessly with one another.
5) *Security Versus Performance:* A tradeoff involving robust security, minimal computational burden, and minimal energy consumption is still an open question today–especially among devices with extremely stringent resource limits. For example, RFID-specific ciphers (Gong et al. [81]) are efficient in situations where high throughput is required.
6) *Real-Time Verification in Dynamic Environments:* Nguyen and Kortun [67] discuss real-time optical ism; while cryptographic verification of any given operation in any given surroundings this could be a building, street signs, or even urban system where people work at different times, must yet be perfected and put forth.

B. Real-World Case Studies

To evaluate the feasibility of lightweight cryptographic algorithms in the context of IIoT, we conducted a series of real-world and laboratory trials to demonstrate their efficacy, efficiency, and usefulness in securing a range of industrial systems. This subsection presents case studies that show the effective deployment of lightweight cryptosystems within different IIoT domains. An overview of these deployments is given in Table VIII in order to show their realistic industry applications as well as to highlight the adopted techniques and performance results.

A lightweight RFID authentication protocol for IIoT nodes operating in a blockchain-based supply chain, where we combine cryptographic hashing with edge-side dynamic key scheduling proposed by Kumar and Das [92]. This system was tested in a logistics chain, which resulted in a sub-5 ms authentication latency with secure asset tracking. The protocol was tailored for resource-constrained RFID and energy-restricted edge computing platforms and was proven compatible with an RFID tag platform by [92].

Jia et al. [93] proposed a lightweight leakage-resilient authenticated key exchange (LLRAKE) scheme for factory automation applications. The protocol relies on hash functions and elliptic curves to provide a secure session setup with low leakage risks. Implementation of LLRAKE demonstrated a memory footprint of less than 2 KB, and good handshake reliability, so as to make LLRAKE promising for deployment in IIoT edge gateways and controllers.

Saheed et al. [94] developed the GA-mADAM-IIoT, a crypto-wrapper mechanism for distributed AD coupled with AI models (attention-based LSTM). This merger delivered secure and intelligent threat detection for streams of IIoT sensors. Field deployment on a multivariate process control system achieved more than 98% accuracy in detecting attacks with energy consumption 28% lower than full-scale encryption approaches.

Li [95] proposed an architecture based on blockchain and CP-ABE for information sharing among diverse IIoT suppliers in smart manufacturing settings. The network secured data privacy and consistency in distributed production nodes, and the processing delay was guaranteed to be less than 500 ms per transaction. It was successful at showing access control enforcement and secure auditability in an actual consortium network.

C. Future Research Directions

To eliminate these gaps, it seems likely that future research will aim for the following goals.

1) *Postquantum Cryptographic Solutions for the IIoT:* Develop cryptographic algorithms that are both light and quantum-resistant, specifically for limited chips. More practically, test their scalability in real-world Industrial Trade scenarios and garment efficiency with Internet of Things techniques.
2) *Innovation in Scalable, Decentralized Authentication Frameworks:* Design scalable network authentication mechanisms for the Industrial Trade that are easy on both compute and power. This is also an approach to deal with the problem of size.
3) *Energy-Aware Hybrid Implementations:* Combine dynamic power scaling with adaptive workload division so as to maximize energy efficiency without sacrificing security features in low-power cryptography applications.
4) *Cross-Platform Standardization:* Establish standards and protocols that can be accepted universally among the diverse Industrial Trade device types and systems, for example, to eliminate differences in power consumption levels.
5) *Context-Based Cryptography Techniques:* Some may involve integrating ML within their physical infrastructure, so senior engineers have a good way of understanding what happens without too much manual intervention
6) *Real-Time Verification Frameworks:* Development verification machines for real-time cryptography in dynamic and latently sensitive ITTE fields, combinative hardware acceleration, and software transference offer a possible way forward.

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TABLE VIII
SUMMARY OF REAL-WORLD LIGHTWEIGHT CRYPTOGRAPHIC CASE STUDIES IN IIOT

Case Study	Authors (Year)	IIoT Application	Lightweight Technique	Deployment Outcome
RFID Authentication for Blockchain Supply Chain	Kumar & Das (2025) [92]	RFID asset tracking in logistics	Hash-based authentication and dynamic keys	Latency $<$ 5 ms, RFID compatibility
LLRAKE Key Exchange Protocol	Jia et al. (2024) [93]	Secure key exchange for factory edge devices	Leakage-resilient key exchange using ECC and hashing	Memory $<$ 2 KB, high handshake success
GA-mADAM-IIoT Threat Detection	Saheed et al. (2025) [94]	Anomaly detection in sensor networks	AI-assisted anomaly detection with encrypted wrappers	98.6% accuracy, 28% energy savings
Blockchain with CP-ABE in Smart Manufacturing	Li (2024) [95]	Secure data sharing in manufacturing networks	CP-ABE for access control with blockchain integration	Latency $<$ 500 ms, audit-ready secure access

7) *Enhanced Privacy for Intermittently Connected Devices:* To provide privacy-protecting protocols that are better able to cope with intermittently connected device states.
8) *Sustainable Cryptographic Solutions:* Use renewable energy solutions(energy harvesting, for example) to maintain the security of Industrial Trade operations over extended periods without access to outside power supplies.

Under the proposed guidelines, addressing these research gaps will put us on the road to potentially stronger, larger, and energy-thrifty-light cryptographic solutions that meet IIoT environments' multifaceted needs. It will definitely get Industry 4.0 into critical operation–it has been confined by test equipment up until now.

VII. NOVEL CONTRIBUTIONS AND EMERGING SECURE IIOT DIRECTIONS

In order to address the gaps identified and further the impact of the survey, we also propose prospective and innovative contributions that can characterize the future secure IIoT systems.

A. Intelligent Adaptive Cryptography for IIoT

The dynamic and diversified characteristics of IIoT systems call for these security mechanisms to be able to accommodate different end-device constraints, communication delays, and dynamic threat scenarios. Lightweight conventional cryptographic mechanisms are particular, since they are mostly static with their settings: once they are rolled out, they use a fixed level of encryption, a certain key length, and a specific algorithmic behavior independently of the runtime use. That inflexibility can waste resources or protect too little.

To tackle this limitation, we introduce a new concept, which is referred to as intelligent adaptive cryptography (IAC) for IIoT systems. This methodology adds a contextual security layer using AI/ML for dynamic contextual parameter tuning in which IIoT devices can be contextually, dynamically, and autonomously tuned to cryptographic parameters based on on-the-fly empirics and constraints.

As shown in Fig. 5, the architecture of the proposed IAC system includes three main parts.

1) *Security Agent Module (SAM):* Constantly observing local parameters, including CPU load, battery state, and detected anomalies, and communication latency.

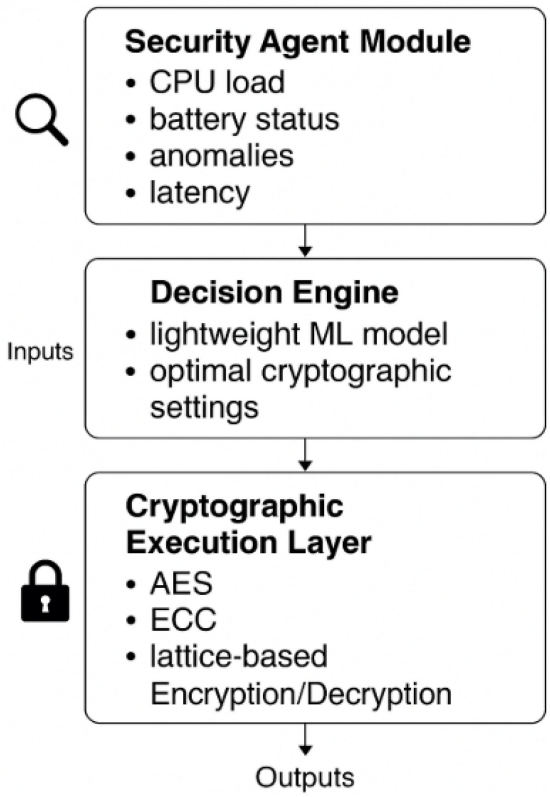

Fig. 5. Architecture of the IAC framework.

2) *Decision Engine (DE):* Simple ML model trained on historical behavior, threat intelligence, and policy objectives. It chooses cryptographic parameters (e.g., key length, encryption mode).
3) *Cryptographic Execution Layer (CEL):* Provides runtime configurations selected using modular components (such as AES, ECC, or lattice-based primitives) and reconfigured without restarting the device.

B. Secure Digital Twin Synchronization

As IIoT systems have matured toward more cyber–physical systems, digital twins (DTs) are one of the core and paramount technologies for predictive maintenance, control, and process optimization. But this synchronization between the physical objects and their digital representation in real-time creates a new attack surface that is not explicitly considered in

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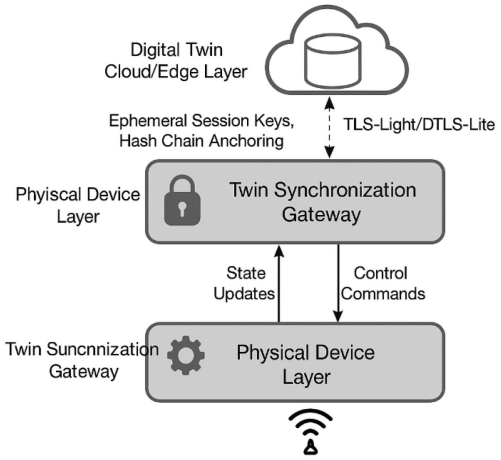

Fig. 6. Secure architecture model for DT synchronization in IIoT.

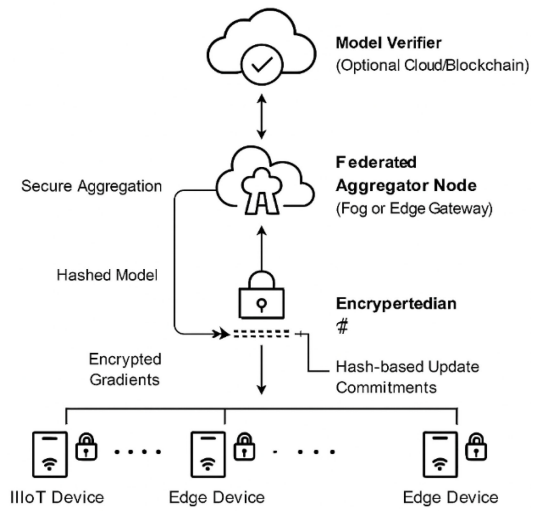

Fig. 7. Secure FL architecture.

traditional IIoT security schemata. The secure synchronization across these two entities requires the development of new cryptography in a lightweight (low overhead, low latency) fashion to provide the required confidentiality, integrity, and authenticity.

As shown in Fig. 6, the secure DT synchronization is achieved by a three-tier architecture.

1) *Physical Device Layer:* Physical layer-mounted IIoT sensors and actuators issue real-time state updates, and process control commands, respectively.
2) *Synchronized Twin Gateway:* Secappears to be a cryptographic protocol middleware: Handles key exchange and rekeying logic. Applies update rate and trust policies. Supports hash-based logs for proving state history.
3) *DT Cloud/Edge Layer*: The DTs of each WSN concurrently communicate to the shared DT in the cloud by the edge server, which closely senses the perception layer. In such a manner, it requires performing the synchronization approach.

Houses the DT, predictive analytics, visualizations, and command generation. TLS-light or DTLS-lite protocols in a joint optimization are used to safeguard communications with the gateway over the constrained channel.

C. Lightweight Federated Learning Security

With the growing dependence of IIoT ecosystems on predictive maintenance, anomaly detection, process optimization, and other ML tasks driven by AI, the new method of FL comes into focus as a decentralized learning paradigm. FL also allows multiple IIoT devices to collectively train ML models without sharing their raw data, which is beneficial for privacy. However, incorporating FL into IIoT also brings distinct security and efficiency challenges in that devices are resource-constrained, suffer from low bandwidth, and have the risk of being manipulated by adversaries. This part presents lightweight cryptographic algorithms proposed for protecting FL in IIoT environments.

As shown in Fig. 7, a reliable FL framework in IIoT consists of the following layers.

1) *Edge Device Level:* At the edge device level, each IIoT node has the capability to train with its own local models on device-level data and to secure the gradients with secure aggregation or DP-based masking.
2) *Federated Aggregator Node (Fog/Edge Gateway):* The fog accepts encrypted updates, aggregates them, checks the update commitments, and broadcasts the global model.
3) *Model Verifier (Optional Cloud/Blockchain):* Provides auditability through hashed log of updates and client participation through smart contracts or a distributed ledger.

D. Green Cryptography for Sustainable IIoT

With the increase in the complexity and number of deployments of Industrial IoT systems, sustainability appears as a cousin requirement of security. Conventional cryptographic algorithms are very exhaustive in terms of energy consumption, generating carbon, and draining batteries, less so in remote or energy-starved locations, and also when deployed in millions of IIoT devices. So, the Green Cryptography paradigm seeks to harmonize the requirement for secure communication with environmental imperatives and also with the duty of giving a more extended lifetime to devices and a minor impact on the environment.

To help IIoT developers, system integrators, and cybersecurity architects in selecting secure and sustainable cryptographic algorithms, this report introduces a three-step framework for sustainable cryptographic decision-making, as shown in Fig. 8.

1) *Security Level Matching:* This first step assesses the level of security needed, determined by the application's risk, data sensitivity, and criticality. High-risk IIoT systems need strong cryptographic measures, but low-risk scenarios can apply for light protection to avoid energy starvation.
2) *Green Scoring Filter:* In this phase, crypto candidate implementations are evaluated under sustainability

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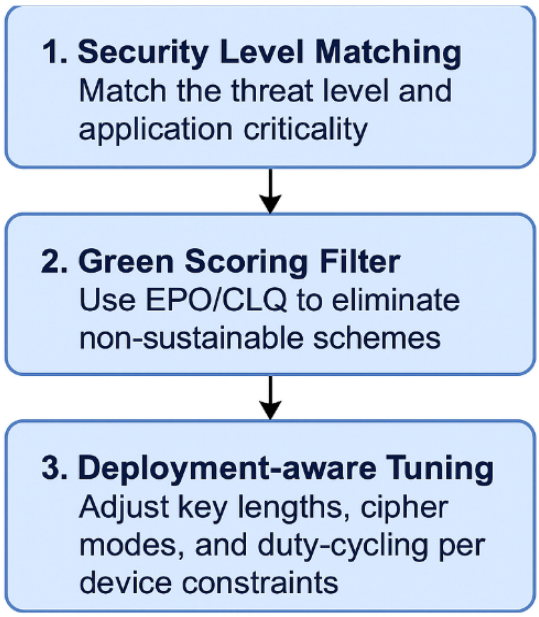

Fig. 8. Framework for sustainable Crypto selection.

metrics, namely, EPO, CLQ, and CCB. Those algorithms that exceed acceptable definitions for energy consumption and emissions are dismissed to respect environmental regulations.

3) *Deployment-Aware Tuning:* The last step is to tailor the chosen cryptographic algorithm for the target IIoT context. This entails tuning the key sizes, choosing energy-efficient cipher modes, and setting the timing of cryptographic operations according to the device's duty cycle, to reduce energy consumption at the cost of security.

VIII. Conclusion

This work provides a systematic, taxonomy-driven review of lightweight cryptographic structures optimized for the security requirements of IIoT settings. Taking all these computation, memory, energy, and real-time operating limitations of IIoT into account, this article presents a five-dimensional taxonomy as to algorithm type; functional objective; implementation domain; application context; and energy efficiency. This classification framework can be utilized as a reference for choosing appropriate types of cryptographic methods across industry applications. The survey also outlines, via comparison and performance evaluation, essential tradeoffs between the computational cost, energy cost, and memory overhead. It also stands out by adopting the most recent cryptographic progress, such as PQC, energy-aware hybrids, and real-time AEAD constructions. Indeed, the work underlines gaps which are: the scarce exploitation of post-quantum schemes in constrained settings, missing or not harmonized solutions across IIoT platforms, and the challenge of trading off latency with strong security properties. To do so, it presents promising future trends such as hybrid cryptographic schemes, scalable and context-aware authentication, and a combination of physical-layer security techniques. By compiling cutting-edge research in a systematic and readable manner, this survey can serve as a roadmap and foundational point for researchers and practitioners intending to develop secure, efficient, and scalable cryptographic designs in the dynamic regime of IIoT.

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